Haotian Zheng

Copenhagen, Denmark | +45 5269 0709 | htzheng@bio.ku.dk | github.com/htzheng1018

EDUCATION

University of Copenhagen — Copenhagen, Denmark
PhD in **Computational Biology** — *Mar. 2026 - Mar. 2029(expected)*
GPA:

University of Michigan Ann Arbor — Ann Arbor, Michigan, US
MS in **Biostatistics** — *Aug. 2023 – May. 2025*
GPA: **3.95/4.0**

Wuhan University — Wuhan, Hubei, China
BS in **Financial Engineering** — *Aug. 2019 – Jun. 2023*
GPA: **3.81/4.0**

- Scholarship for Academic Excellence, WHU — *Nov. 2021*
- Merit Student Award, WHU, Twice — *Nov. 2020 & Nov. 2021*

RESEARCH EXPERIENCE

Department of Computational Biology, SarahRennie Lab, University of Copenhagen — *Sep. 2025 - Mar. 2026*
Research Assistant

- Conducted comprehensive genomic analysis in Arabidopsis thaliana linking m6A modification stoichiometry with GWAS variants across 11 custom genomic intervals using GRanges and Fisher's exact tests;
- Analyzed the mechanism of A-to-I RNA editing perturbed in Parkinson's Disease using bulk RNA-seq data and scRNA-seq data from FOUNDINPD project;
- Developed high-throughput parallel R pipelines by *doParallel* to implement Generalized Linear Mixed Models for over 30,000 editing sites, quantifying significant different editing in PD patients over time.

Department of Biostatistics & Bioinformatics, AviKenny Lab, Duke University — *Jun. 2024 - present*
Graduate Research Assistant

- Utilized the *SimEngine* package in R to simulate Cox model datasets within a two-phase study design, optimized algorithms by implementing parallelization to improve computational efficiency;
- Derived mathematical expressions for estimators, including the expectation of the Survival Function for both placebo and treatment groups, in a Controlled Vaccine Efficacy model with biomarkers, using Breslow estimators and partial likelihood methods;
- Building upon the derived estimator expressions, extended the model to incorporate a two-phase sampling design, integrating inverse-probability weighting (IPW) to accommodate the new framework, and subsequently derived the mathematical expressions for the new second-phase estimator;
- Developed an R program to verify the accuracy of the aforementioned estimators, used Bootstrap algorithms to calculate the standard errors, and generated visualizations to compare the new estimator with the true survival function. Evaluated the efficiency of the new second-phase estimators compared to Kaplan-Meier estimators;
- Validated model performance by deriving the asymptotic variance of estimators using influence functions and the delta method, enabling the construction of valid confidence intervals;
- GitHub link: github.com/htzheng1018/sim_surv_model.

Department of Biostatistics, Epidemiology and Informatics, DaweiXie Lab, University of Pennsylvania — *Jun. 2024 - Feb. 2025*
Graduate Research Assistant

- Utilized the *mice* package in R to analyze and impute longitudinal RAND HRS survey data;
- Designed a prediction matrix to leverage predictor-specific information for imputation of Composition data and Interactive data;
- Developed a function to align datasets based on their density characteristics, enabling effective distributional correspondence;
- Created functions to rationalize imputation algorithms to accommodate real-world conditions, such as maintaining logical consistency in delayed vs. immediate recall scores;
- Introduced methods for stratified imputation, integrating multiple imputation strategies tailored separately to specific conditions;
- GitHub link: github.com/htzheng1018/longitudinal_imputation.

Department of Biostatistics, AlexanderTsodikov Lab, University of Michigan — *Nov. 2023 - Jan. 2025*
Reading

- Studied Proportional Hazard Frailty Models (PHFM) under missing data conditions;
- Implemented EM and MM algorithms for model estimation, comparing the relative efficiency of different estimations, include traditional MLE, weighted Breslow estimators;
- Analyzed the application of Martingale and Submartingale theory in the PHFM estimation process;

- Researched advanced statistical topics including Copula models, Partially Observed Marker Processes, Lévy Processes, and Pseudo-observations.

Department of Economics and Management, ZhongkuangZhao Lab, Wuhan University *Jul. 2021 - Jun. 2023*
Undergraduate Research Assistant
- Analyzed capital squeeze and economic activities from the perspectives of balance sheet and bank credit, focusing on the effects of entrepreneurial ability and supervision channels on borrowing activities.

Sloan School of Management, EgorMatveyev Lab, MIT *Jul. 2021 - Oct. 2021*
Undergraduate Research Assistant
- Calculated and analyzed key financial indicators (MV, EV, P/E Ratio, EV/EBITDA, EPS) for over 5,000 stocks;
- Identified a high-potential company through quantitative screening and fundamental analysis, culminating in a investment report detailing its growth prospects and reasons for investment.

PUBLICATIONS

Trends in Trajectories and Incidence of Functional Impairment in Middle-Aged Americans, Journal of General Internal Medicine, under review Oct 2025
Rebecca T. Brown, Xi Zuo, ***Haotian Zheng***, Kira Ryskina, Anne R. Cappola, Irma Elo, Norma B. Coe, Dawei Xie

INTERNSHIP EXPERIENCE

Lingjun Investment, LLP, Beijing, China *Jun. 2022 - Sep. 2022*
Quantitative Research Intern
- Designed and implemented an automated pipeline in Python for factor discovery, backtesting, and correlation analysis, yielding over 300 high-performing alpha factors with low correlation (<60%) to existing libraries, spanning signals such as price-volume, momentum, reversal, and fundamentals;
- Enhanced the performance of out-of-sample validated factors through techniques including weighted factor values, industry neutralization, and market-neutral strategies;
- Developed C++ operators for high-frequency data computation, including moving averages, maximum values, and weighted averages, while handling missing values;
- Utilized Linux for batch backtesting, factor correlation analysis, and filtering of optimal factors;
- Reproduced and enhanced quantitative factors from A-share research reports, tailoring them to the unique characteristics of the Hong Kong stock market to achieve excellent backtested performance over a ten-year period.

Techfin.ai Co., Ltd., Beijing, China *Jan. 2022 - Apr. 2022*
Intern at Algorithm Dept.
- Researched and reproduced established quantitative factors from academic literature using Python (numpy, pandas), then developed novel factors based on daily high-frequency A-share trading data;
- Validated these factors by calculating their Information Coefficient (IC) and correlation, successfully identifying 28 effective signals with an IC greater than 3% and low correlation to existing factors.

China Securities Co., Ltd., Remote *Jan. 2022 - Apr. 2022*
Personal Assistant to an Analyst
- Built Python web crawlers for stock data collection and preprocessing, including mean imputation, normalization, and outlier removal;
- Constructed and validated a momentum factor, achieving an IC of 3.6%;
- Analyzed the performance of snowball options and implemented a quantitative timing trading strategy.

TEACHING EXPERIENCE

Department of Computational Biology, University of Copenhagen *Feb. 2026 - Apr. 2026*
Teaching Assistant of xxxxxx

Department of Biostatistics, University of Michigan *Sep. 2024 - Apr. 2025*
Facilitator of study group in BIOSTAT 651 – Theory and Application of Generalized Linear Models
Facilitator of study group in BIOSTAT 650 – Theory and Application of Linear Regression

Department of Economics and Management, Wuhan University *Feb. 2023 - Jun. 2023*
Teaching Assistant in Dynamic Optimization

SKILLS

Programming: R, Python, C++, C, Shell
Toolkits: Git, Latex
Languages: Chinese, English, Danish
Hobbies: Soccer, Badminton, Piano